

On the Security of a Code-Based PIR Scheme

WCC 2026, Paris

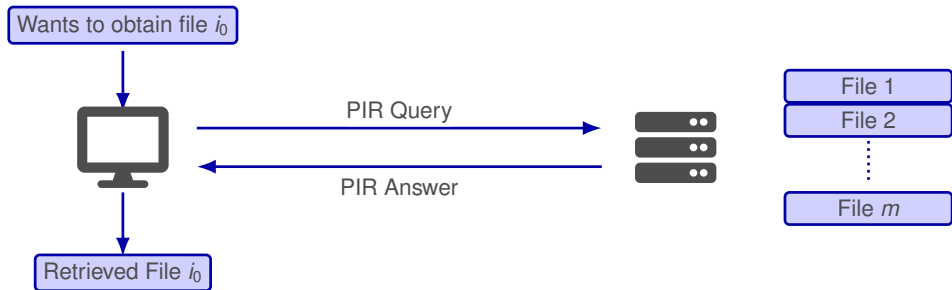
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Private Information Retrieval (PIR)



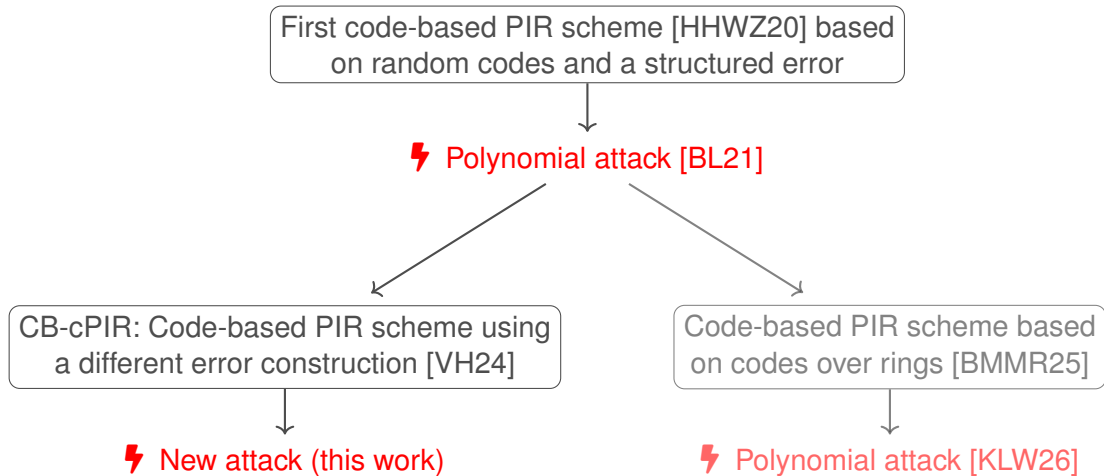
- **Privacy:** Server learns nothing about index i_0

Information-Theoretic Security

- Single Server: Only trivial solution (download entire database)
- Multiple Servers: Efficient solutions exist under the assumption of **non-colluding** servers

Computational Security

- Possible in the single-server scenario
- Most post-quantum schemes rely on lattice-based cryptography
- Alternative assumptions (e.g. code-based approaches) are desirable



Outline



1. The CB-cPIR scheme

2. Attacking the scheme

3. Outlook

Setup:

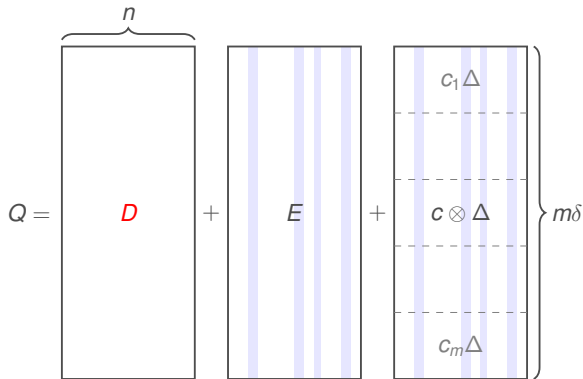
- Database containing m files $X^1, \dots, X^m$
- Each file is represented as $X^i \in \mathbb{F}_q^{L \times \delta}$
- Database $X \in \mathbb{F}_q^{L \times m\delta}$

$$X = \left[\begin{array}{|c|c|} \hline X^1 & X^2 \\ \hline \end{array} \right] \overset{\delta}{\brace{}} \dots \left[X^m \right] \Bigg\} L$$

- Goal: Privately retrieve file X^{i_0}

Query generation:

- Let $\delta = (n - k)(s - v)$
- Choose a random $[n, k]$ -linear code $\mathcal{C} \subseteq \mathbb{F}_{q^s}^n$
- Construct $D \in \mathbb{F}_{q^s}^{m\delta \times n}$ whose rows are codewords of $\mathcal{C}$
- Choose an information set I with $|I| = k$

$$Q = \overbrace{\begin{matrix} n \\ \boxed{D} \end{matrix}} + \begin{matrix} \boxed{E} \end{matrix} + \left. \begin{matrix} \begin{matrix} c_1 \Delta \\ \hline c \otimes \Delta \\ \hline c_m \Delta \end{matrix} \end{matrix} \right\} m\delta$$
The diagram illustrates the query generation process. It shows three matrices being summed to form the query Q. The first matrix is labeled D and has a width of n, indicated by a brace above it. The second matrix is labeled E and has several vertical blue stripes. The third matrix is grouped by a large right curly brace labeled mδ. This matrix contains rows labeled c1Δ, c ⊗ Δ, and cmΔ, with vertical blue stripes. The matrices are separated by plus signs.

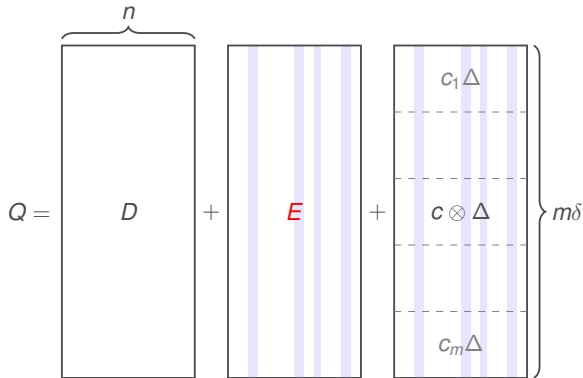
Query generation:

- Choose a random basis

$$\Gamma = \{\gamma_1, \dots, \gamma_s\}$$

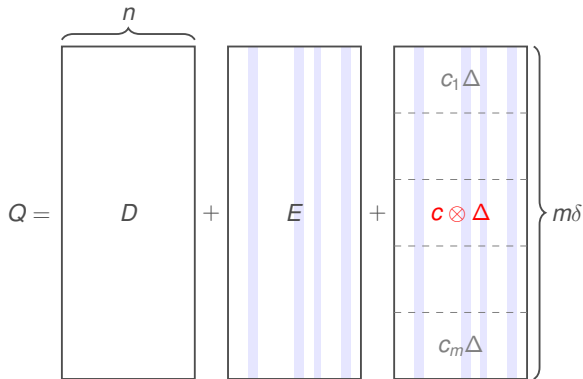
of $\mathbb{F}_{q^s}$ over $\mathbb{F}_q$

- Let $V = \langle \gamma_1, \dots, \gamma_v \rangle$ and $W = \langle \gamma_{v+1}, \dots, \gamma_s \rangle$
- Construct E with entries from V outside I and zeros elsewhere

$$Q = \overbrace{\begin{matrix} n \\ D \end{matrix}} + \begin{matrix} E \end{matrix} + \left. \begin{matrix} c_1 \Delta \\ \vdots \\ c \otimes \Delta \\ \vdots \\ c_m \Delta \end{matrix} \right\} m\delta$$


Query generation:

- Randomly sample $\beta \in (\mathbb{F}_q^\times)^m$
- Set $c = \beta + e_{i_0}$
- Construct $\Delta \in W^{\delta \times n}$ with entries from W outside I and zeros elsewhere
- Calculate $c \otimes \Delta$

$$Q = \overbrace{\begin{matrix} n \\ D \end{matrix}} + \begin{matrix} E \end{matrix} + \left. \begin{matrix} c_1 \Delta \\ \vdots \\ c \otimes \Delta \\ \vdots \\ c_m \Delta \end{matrix} \right\} m\delta$$
The diagram illustrates the query generation process. It shows three matrices being summed to form the query Q. The first matrix, labeled D, has a width of n. The second matrix, labeled E, has vertical blue stripes. The third matrix is a block matrix with m rows, each representing a vector c_i multiplied by a matrix Delta. The middle row is highlighted with red text 'c \otimes \Delta'. The entire third matrix is grouped by a brace on the right labeled m\delta.

The CB-cPIR scheme

Query generation:

- A second matrix is required for file recovery
- Repeat the process to generate another matrix Q_β but use the same β

$$Q = \begin{matrix} \boxed{D} & + & \boxed{E} & + & \begin{matrix} c_1 \Delta \\ \vdots \\ c \otimes \Delta \\ \vdots \\ c_m \Delta \end{matrix} \end{matrix} \qquad Q_\beta = \begin{matrix} \boxed{D_\beta} & + & \boxed{E_\beta} & + & \begin{matrix} \beta_1 \Delta_\beta \\ \vdots \\ \beta \otimes \Delta_\beta \\ \vdots \\ \beta_m \Delta_\beta \end{matrix} \end{matrix}$$

$$c = \beta + e_{i_0}$$

- Send the query (Q, Q_β) to the server

Answer generation:

- The server calculates the answer $R = (X \cdot Q, X \cdot Q_\beta) \in \mathbb{F}_{q^s}^{L \times n} \times \mathbb{F}_{q^s}^{L \times n}$

File extraction (high level):

- The client removes codeword components using information set decoding
- The error components E and E_β are removed by projection
- Combining both parts of the response, the client can recover the desired file

Outline



1. The CB-cPIR scheme

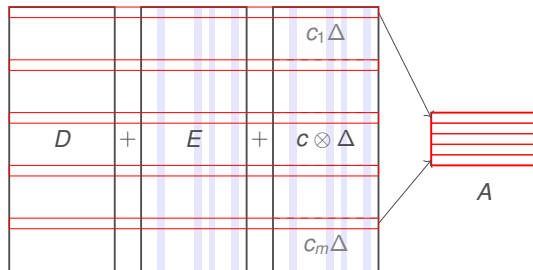
2. Attacking the scheme

3. Outlook

Attacking the scheme

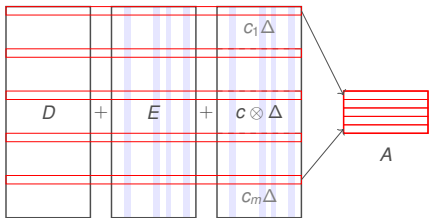
Step 1: Construct an auxiliary matrix

- Construct $A \in \mathbb{F}_{q^s}^{m \times n}$ by extracting the first row of every block of Q



- While $\text{rk}_{\mathbb{F}_q}(Q) = ns$, we can show that

$$\text{rk}_{\mathbb{F}_q}(A) \leq ks + (n - k)v + 1 = ns - \delta + 1$$



- A already spans the code $\mathcal{C}$
- Appending codewords to A does not increase the rank
- Appending rows of E does not increase the rank
- Appending a different row of Δ increases the rank

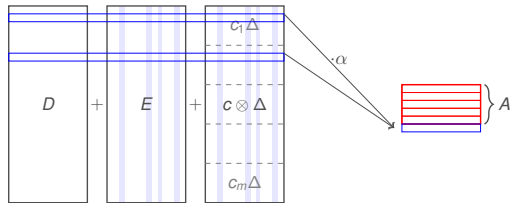
$\Rightarrow$ We can detect the presence of Δ

Step 2: Recover relations between the coefficients of c

- Start with blocks one and two
- For $\alpha \in \mathbb{F}_q^\times$, append $\alpha Q_{[2,:]}^1 + Q_{[2,:]}^2$ to A
- Rank test:

$$\alpha c_1 + c_2 = 0 \Rightarrow \text{rank unchanged}$$

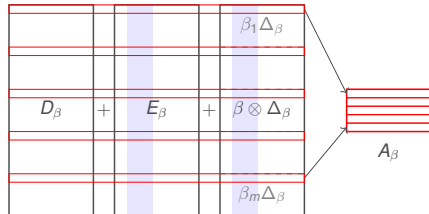
$$\alpha c_1 + c_2 \neq 0 \Rightarrow \text{rank increases}$$



- Iterating over $\alpha \in \mathbb{F}_q^\times$ reveals the relation $\alpha c_1 + c_2 = 0$

Step 3: Check for the index of interest

- Repeat step 1 on Q_β to obtain A_β



- Append $\alpha Q_{\beta[2,:]}^1 + Q_{\beta[2,:]}^2$ to A_β
- The rank test reveals if $\alpha\beta_1 + \beta_2 = 0$

- Recall that $c = \beta + e_{i_0}$
- If

$$\alpha c_1 + c_2 = 0 = \alpha \beta_1 + \beta_2,$$

then $i_0 \neq \{1, 2\}$

- If

$$\alpha c_1 + c_2 = 0 \neq \alpha \beta_1 + \beta_2,$$

then $i_0 \in \{1, 2\}$

- The attack reconstructs the index of interest with a runtime of $O(\lceil \frac{q}{\delta-1} \rceil)$
- Attack succeeds w.h.p. once the database contains sufficiently many files
- Supports parallel execution

q	s	v	n	k	δ	Security level	Attack complexity	Minimal value of m
2^{16}	12	10	100	50	100	2^{113}	2^9	1101
$2^{32} - 5$	6	4	120	60	120	2^{128}	2^{25}	601
2^{32}	5	3	100	50	100	2^{96}	2^{25}	401
$2^{61} - 1$	6	2	100	50	200	2^{113}	2^{53}	401

Outline



1. The CB-cPIR scheme

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- Following our preprint, the authors proposed a revised version of the scheme [HV25]
- We subsequently developed a more general attack that breaks all variants of the scheme
- The new attack is even more efficient and applies to all parameter choices. Consequently, all proposed parameter choices are insecure
- The advanced attack will be published in the full version of the paper

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- [VH24] [N. Verma and C. Hollanti.](#)
Code-based single-server private information retrieval: Circumventing the sub-query attack.
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